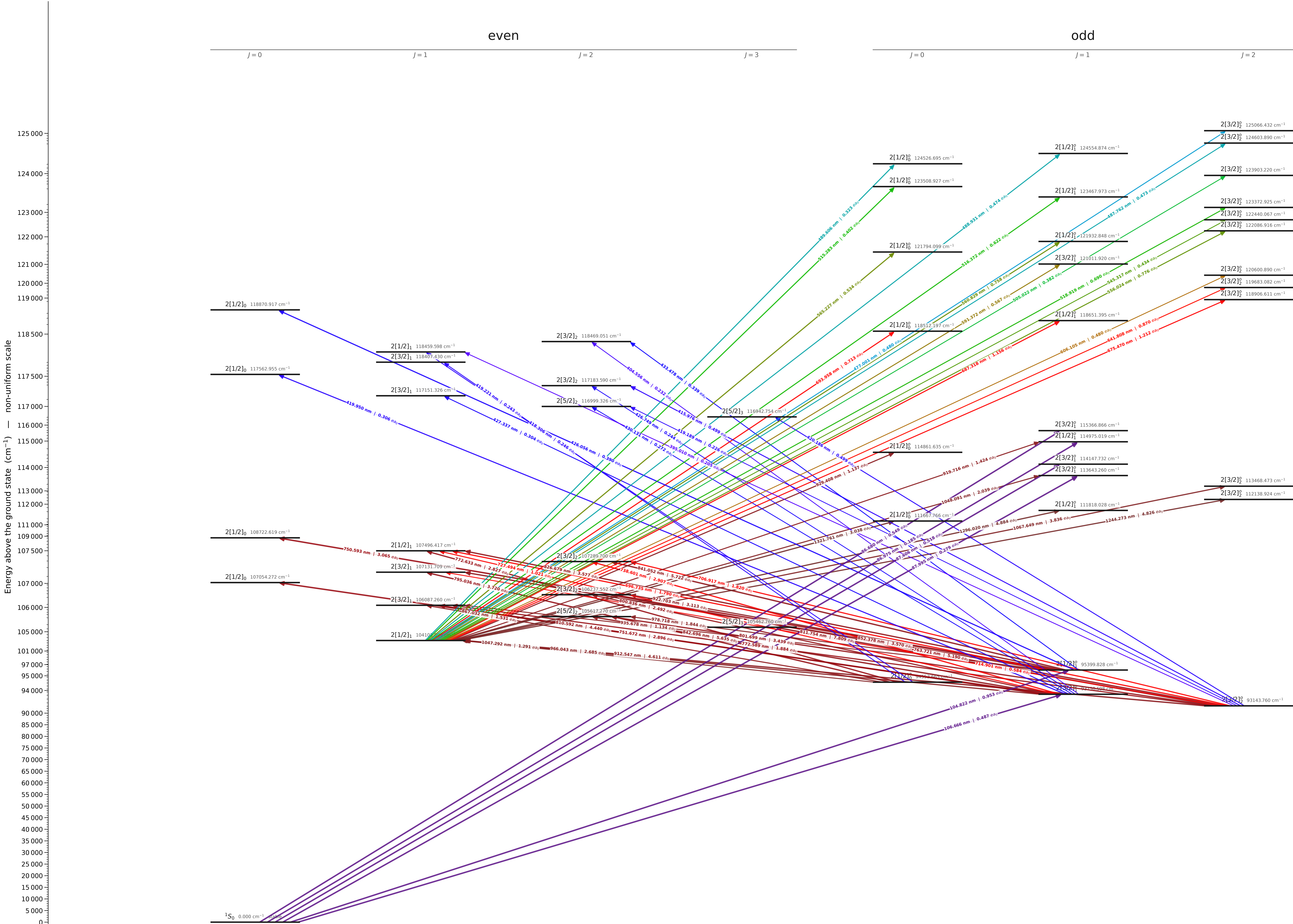
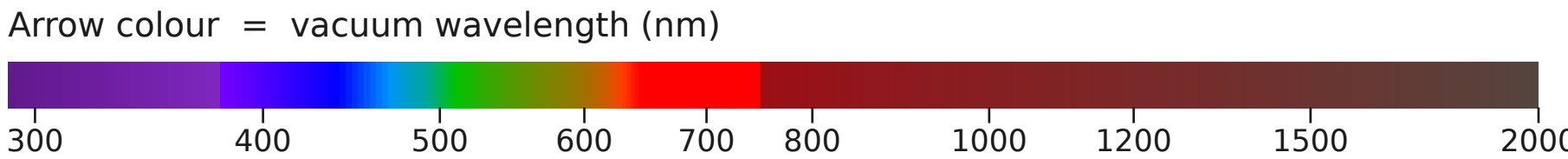


Argon

Ar (Ar I). Strongest classified transitions below 2 μm (NIST): 51 levels, 70 lines.

Ar⁺ ionisation limit 127109.84 cm⁻¹ = 15.75961 eV (78.672 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle U|er||J\rangle|$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U er J\rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ - 2[3/2] ^o ₁	86.6800	-	3458.6116	0.549	3.130e+08	-	-	-
¹ S ₀ - 2[1/2] ^o ₁	86.9754	-	3446.8644	0.185	3.500e+07	-	-	-
¹ S ₀ - 2[3/2] ^o ₂	87.6058	-	3422.0629	0.518	2.700e+08	-	-	-
¹ S ₀ - 2[3/2] ^o ₂	87.9947	-	3406.9392	0.279	7.700e+07	-	-	-
¹ S ₀ - 2[1/2] ^o ₂	104.8220	-	2860.0149	0.953	5.320e+08	-	-	-
¹ S ₀ - 2[3/2] ^o ₂	106.6660	-	2810.5722	0.487	1.320e+08	-	-	-
2[1/2] ^o ₁ - 2[1/2] ₀	750.5934	750.3868	399.4072	3.065	4.500e+07	-	-	-
2[3/2] ^o ₁ - 2[1/2] ₀	751.6720	751.4651	398.8341	2.896	4.000e+07	-	-	-
2[3/2] ^o ₂ - 2[3/2] ₁	763.7207	763.5105	392.5420	5.190	2.450e+07	-	-	-
2[1/2] ^o ₂ - 2[3/2] ₁	795.0362	794.8176	377.0802	3.720	1.860e+07	-	-	-
2[3/2] ^o ₂ - 2[3/2] ₁	810.5920	810.3692	369.8438	4.440	2.500e+07	-	-	-
2[3/2] ^o ₂ - 2[5/2] ₃	811.7542	811.5311	369.3143	7.809	3.300e+07	-	-	-
2[1/2] ^o ₂ - 2[1/2] ₁	826.6793	826.4521	362.6466	3.577	1.530e+07	-	-	-
2[1/2] ^o ₂ - 2[3/2] ₂	841.0519	840.8208	356.4494	5.722	2.230e+07	-	-	-
2[3/2] ^o ₂ - 2[5/2] ₂	842.6962	842.4647	355.7539	5.635	2.150e+07	-	-	-
2[3/2] ^o ₂ - 2[1/2] ₂	912.5471	912.2966	328.5227	4.611	1.890e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2n: natural linewidth of the upper level, 1/(2πn).